Practice workbook · Does a geometric series have a sum to infinity?

Stating or showing that a geometric series has a sum to infinity because $|r| < 1$, or has none because $|r| > 1$.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-12

1. Question 7 · part b · 2 marks

2020 NOV · Paper 02

Worked-example retry · Attempt part b only.

7 A geometric series has first term $(x - 3)$, second term $(x + 1)$ and third term $(4x - 2)$.

(a) Find the two possible values of x.

(5)

Given that $x < 1$,

(b) show that the series is convergent.

(2)

The sum to infinity of the series is S.

(c) Find the value of S.

(2)

The sum of the first n terms of the series is S_n

Given that $\frac{S}{S_n} = \frac{256}{255}$

(d) find the value of n.

(3)

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1. Question 7 · part b · 2 marks · Working and answer

Working and answer

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2. Question 4 · part b · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part b only.

4 A geometric series G has first term a and common ratio r

The third term of G is 10 and the seventh term of G is 40

Given that the second term of G is negative,

(a) find

(i) the exact value of r

(ii) the value of a

(5)

(b) Explain why G does not have a sum to infinity.

(1)

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2. Question 4 · part b · 1 mark

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Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 6 marks)

2. Question 4 · part b · 1 mark · Working and answer

Working and answer

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3. Question 8 · part b · 1 mark

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

8 The sum of the first 2 terms of a geometric series G is 360
The sum of the 2nd and 3rd terms of G is 288

The nth term of G is U_n

(a) Show that $U_n = A\left(\frac{4}{5}\right)^{n-1}$ where A is an integer to be found.

(7)

(b) Explain why G is convergent.

(1)

(c) Hence find the sum to infinity of G

(2)

(d) Find the least number of terms for which the sum is greater than 978

(4)

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3. Question 8 · part b · 1 mark · Working and answer

Working and answer

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4. Question 6 · part b · 1 mark

2025 NOV · Paper 02

Further practice · Attempt part b only.

6 (i) A geometric series G has first term $\left(3+\sqrt{3}\right)$ and common ratio r

The second term of G is $\sqrt{3}$

(a) Find, showing all your working, the exact value of r

Give your answer in the form $\frac{p+\sqrt{3}}{q}$ where p and q are integers to be found.

(2)

(b) Explain why G is convergent.

(1)

(ii) A different geometric series H has first term 8 and common ratio 0.6

The sum to n terms of H is S_n and the sum to infinity of H is T

Find, using logarithms, the least value of n such that $T - S_n < 0.12$

(5)

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4. Question 6 · part b · 1 mark · Working and answer

Working and answer

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